Role of video review for sport-related concussion identification: a systematic review

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ABSTRACT

Objective To determine the frequency and diagnostic utility of visible signs identified through video review for sport-related concussion detection and to evaluate the role of injury mechanism in predicting concussions.

Design Systematic review.

Data sources MEDLINE, SPORTDiscus, Web of Science and Cochrane Library from 2001 to June 2024.

Eligibility criteria Original, peer-reviewed studies focusing on sport-related concussion, where video review identified visible signs or the mechanism of injury relevant to clinical decision-making. Studies were included irrespective of study design but had to be published in English and use a clinical diagnosis or consensus definition of sport-related concussion as a reference standard.

Results Out of 1001 records screened, 29 studies involving 3281 sport-related concussions were included for quantitative synthesis. Only eight studies had a low risk of bias. All studies used clinical diagnosis as a reference standard; 26 were conducted in professional competitions, with two studies including female athletes. The most frequently observed visible sign was 'slow to get up' (37.6%), followed by 'motor incoordination' (26.7%). Sensitivity of visible signs was generally low (<50%), while specificity was high (>90%) across most studies. Four studies across three sports examined mechanism of injury characteristics, identifying common themes, such as high speed of impact, potentially associated with sport-related concussion occurrence. Inter-rater and intrarater reliability varied, with indications of higher reliability among expert raters.

Conclusions Expert video review can reliably identify visible signs of sport-related concussion. However, due to their low frequency and limited diagnostic sensitivity, the diagnosis of sport-related concussion should be supported by a multimodal assessment.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Sport-related concussion is a challenging clinical diagnosis, emphasising the importance of early identification to mitigate potential adverse outcomes and guide management.
⇒ Overt signs of sport-related concussion, such as loss of consciousness, are reported in less than 10% of cases.
⇒ The combined approach of video review and multimodal assessment has shown benefits in aiding the clinical diagnosis of sport-related concussion across various contact sports. However, an evidential summary of the findings and utility of video review for the diagnosis of sport-related concussion has not been performed.

WHAT THIS STUDY ADDS

⇒ Among the 29 studies meeting inclusion criteria, 6 studies (718 sport-related concussions) evaluated the diagnostic accuracy of individual visible signs. Most visible signs showed low sensitivity (<50%) and high specificity (>90%).
⇒ 10 studies reported on the occurrence of visible signs using similar definitions. The two most frequently observed visible signs were 'slow to get up' (37.6%) and 'motor incoordination' (26.7%). This was followed by 'no visible sign seen' in almost a quarter of diagnosed concussions (23.6%).
⇒ Given the low frequency and sensitivity of visible signs, this adjunctive tool should be complemented by a multimodal assessment for the diagnosis of sport-related concussion. Future studies should address methodological limitations, including the inclusion of female athletes and consider how sport-specific mechanisms of injury influence risk-prediction models.

INTRODUCTION

The assessment and management of athletes who have sustained a sport-related concussion has developed significantly in recent years, supported by an international consensus definition and frequent systematic reviews of scientific literature guiding research efforts.[1] Despite these advancements, sport-related concussion remains a challenging clinical diagnosis. Early identification is crucial to mitigate potential adverse outcomes and facilitate timely assessment before implementing appropriate management strategies for a safe return to play.[2] Identifying sport-related concussion early is challenging as overt signs, such as loss of consciousness, are present in less than 10% of cases.[3] Moreover, beyond the infrequent occurrence of visible signs, additional challenges include the fast-paced and often chaotic environment of elite-level sports, where medical staff may have obscured views or be distracted by injuries elsewhere on the field.[2]

In the absence of a definitive pitch-side biomarker for identifying sport-related concussion, the recognition of suspected cases is best approached using a multimodal side-line assessment tool like the Sport Concussion Assessment Tool (SCAT), complemented by video review capabilities.[2] When available, video review allows clinicians to review

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incidents in detail, potentially capturing brief early signs of sport-related concussion following impact. Consequently, many professional contact sports have incorporated video review into their sport-related concussion management protocols.[4] The adoption of video review processes varies across sports, influenced more by resource availability and sport-specific demands rather than strictly scientific evidence.[4] However, the effectiveness of video review is influenced by the availability and quality of footage, which is often limited at the non-professional level due to fewer camera angles and lower resolution.[5]

Nevertheless, the combination of video and SCAT-based assessments has shown benefits in diagnosing sport-related concussion in various contact sports.[6 7] Despite differing practical protocols, there are commonalities across sports in the visible signs examined through video review.[4] International consensus definitions of six primary concussion video signs have provided clarity and promoted consistent research, with the objective of enhancing match-day detection and standardised recognition practices.[8]

This systematic review aims to evaluate the existing body of evidence on:

- The prevalence of visible signs observed through video review in sport-related concussions.
- Inter-rater reliability in identifying visible signs of sport-related concussion via video review.
- The diagnostic accuracy of visible signs identified in sport-related concussion cases.
- The role of injury mechanisms in predicting sport-related concussion.

This review includes original research articles from various sports and observers with diverse levels of experience.

METHODS

Study design and reporting

This systematic review follows the evidence-based minimum standards outlined in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses of Diagnostic Test Accuracy (PRISMA) (online supplemental eFigure 1) and implementing PRISMA in Exercise, Rehabilitation, Sports medicine and SporTs science (PERSiST) guidelines.[9 10] The review was not preregistered prior to it being conducted.

Eligibility criteria

Studies eligible for inclusion had to be peer-reviewed, written in English, provide full-text articles (not abstract only), report ethical approval and participant consent where applicable, and meet the eligibility criteria (online supplemental eTable 1). In this systematic review, the primary focus was on concussion sustained during sports competition or training (sport-related concussion), where video review was used to identify visible signs or mechanism of injury relevant to clinical decision-making. Studies were required to employ either a clinical diagnosis by a medical doctor or apply a consensus definition of sport-related concussion as the reference standard for diagnosing sport-related concussions.

Search strategy

The Population, Index test, Target condition framework guided the identification of key topics for inclusion in the search (online supplemental eTable 2).[11] Synonymous search terms related to these key topics were identified to ensure comprehensive coverage (online supplemental eTable 3). To enhance sensitivity, additional search terms were incorporated after screening titles, abstracts and full texts of relevant studies initially identified. These terms were trialled in pilot searches before integration into the final search strategy.

The inclusion start date was set as 1 January 2001, aligning with the establishment of the first consensus definition of sport-related concussion.[12] Searches were independently conducted on MEDLINE, SPORTDiscus, Web of Science and Cochrane Library from 1 November 2022 to 30 November 2022 by the principal reviewers (DCT and AS). To maintain currency, an updated search was conducted up to and including June 2024. Boolean search command "OR" was used to combine the main concept with associated synonyms, and the command "AND" was used to include all three main concepts within the search. The "*" symbol was used for wildcard character searching as needed.

Screening strategy and study selection

All citations identified by the search strategy were managed in EndNote (Clarivate Analytics, Philadelphia, Pennsylvania, USA). Duplicate studies were identified and removed. Screening initially involved reviewing titles and abstracts. If inclusion could not be determined from these alone, the full text was consulted briefly. Two reviewers (DCT and AS) independently performed duplicate removal and initial screening. Any disagreements on inclusion were resolved through discussion. If consensus could not be reached, a third reviewer (AB) made the final decision.

Full texts of all the remaining articles underwent a second screening phase. Two reviewers (DCT and AS) independently assessed each article against the eligibility criteria to ensure correct inclusion decisions. Any discrepancies were resolved through discussion or by consulting the third reviewer (AB) as necessary.

The study selection process is visually represented in the PRISMA flow chart (figure 1).

Data extraction

Data from the selected studies were extracted into an Excel spreadsheet by the lead reviewer (DCT). Extracted data encompassed methodological characteristics, participant demographics, study characteristics and study findings (table 1). For this review, 'expert' raters were defined as personnel with relevant prior experience or training in video review, otherwise, they were defined as 'novice' raters.

A second reviewer (AS) independently verified the data extraction process. Any discrepancies were resolved through consensus or consultation with a third reviewer (AB). Attempts were made to obtain missing data by contacting study authors via email.

Risk of bias assessment

Each reviewer (DCT and AS) independently performed a risk of bias assessment, with discrepancies resolved by a third reviewer (AB). Risk of bias was assessed using the Quality Assessment of Diagnostic Accuracy Studies, version 2 (QUADAS-2) framework.[13] Studies were classified as having a 'low risk of bias' if they were rated as 'low' in all domains. Studies were considered 'at risk of bias' or had 'concerns regarding applicability' if one or more domains were rated as 'high' or 'unclear'.[13]

Statistical analysis

Data synthesis and statistical analysis were performed based on available study data. Results are mostly presented descriptively.

For diagnostic test accuracy data, forest plots were generated to illustrate individual study estimates of sensitivity and

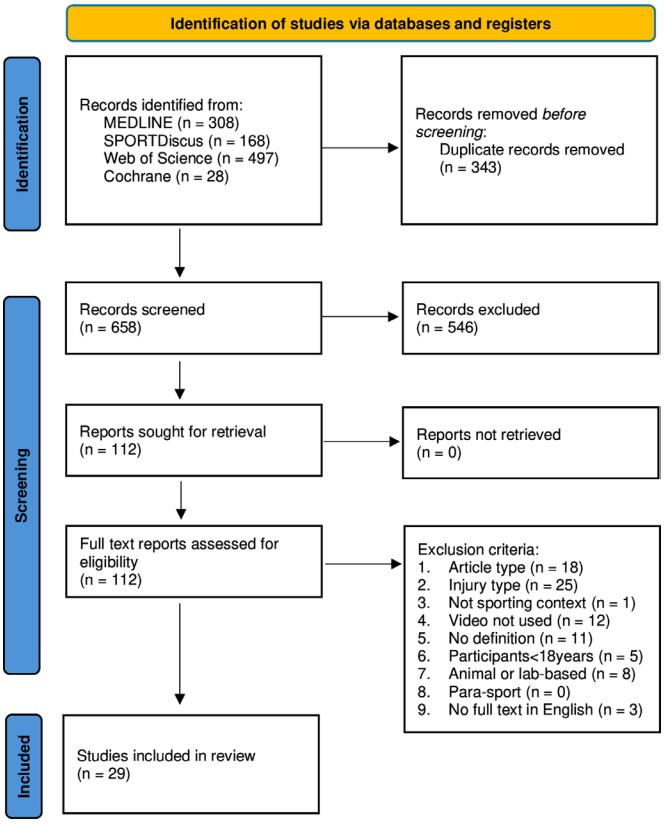


Figure 1 PRISMA flow chart for the screening process. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

specificity for each of the visible signs of sport-related concussion (the index tests), along with 95% CIs. A bivariate random-effects logistic regression model was fitted for each index test separately using the 'meqrlogit' command in Stata V.16.0 to estimate summary sensitivity and specificity values, as recommended in the Cochrane handbook for systematic reviews of diagnostic test accuracy.[14] We planned to conduct a comparative meta-analysis to compare index tests with a summary sensitivity and specificity greater than 50%. All analyses were conducted using RevMan V.5.4 and Stata V.16.0.

Table 1 Data extraction details by category

Category	Data extraction details
Methodological characteristics	► Study aims and design ► Level of evidence ► Study environment (including which sport the study was conducted in)
Participant demographics	► Gender ► Age ► Participant number ► Playing position ► Standard of competition
Study characteristics	► Concussion: definition, method of case confirmation ► Video analysis: technology used, application of analysis, sign description, sign occurrence, mechanism of injury characteristics, sensitivity and specificity, likelihood ratios, diagnostic ORs, predictive models, receiver operating characteristic curve data ► Rater: level of experience, intrarater and inter-rater reliability ► Outcome measures: diagnosis, player removal from competition/training, duration of absence, return to play progression, postconcussion syndrome
Study findings	► Main results ► Limitations

Equity, diversity and inclusion statement

All eligible studies were included in this review without exclusion based on participant characteristics such as sex, gender, ethnicity, socioeconomic and competition level. The discussion section addresses the importance of enhancing diversity within the evidence base on this topic.

The research team comprises individuals of diverse backgrounds, including men and women from various educational backgrounds, professional disciplines and levels.

RESULTS

Study selection and characteristics

The initial search across electronic databases yielded 1001 articles. After removing duplicates and applying eligibility criteria, 29 studies were included (figure 1) with a combined total of 3281 sport-related concussions undergoing video review. From the information available in the studies, the sport-related concussion cases included in them are not known to overlap. Table 2 presents a summary of key study characteristics.

Among the included studies, 27 focused exclusively on male athletes, one on female athletes[15] and one compared both sexes.[16] All studies used a clinical sport-related concussion diagnosis as the reference standard. 15 studies also reviewed head impact events that did not result in a clinical diagnosis (n=22 076) in addition to the sport-related concussion cases; this was primarily for comparison purposes.[15 17–30] The majority of studies were retrospective (n=19) and performed in professional sporting competitions (n=26). The number of concussion cases varied greatly (ranging from 0 to 429), as did the framework used for video analysis—most commonly performed by expert raters (n=19).

10 studies exclusively analysed the mechanism of injury through video review,[15–17 19 30–35] while 11 studies focused solely on visible signs,[4 8 20–24 26–28 36] and 8 studies examined both elements in combination.[18 25 29 37–41]

Risk of bias and applicability

On review, eight studies were categorised as having a 'low' risk of bias,[4 8 20 23 26–28 41] while the remaining studies were either deemed 'at risk of bias' (n=11)[15–17 19 24 25 31 32 34 35 39] or had 'concerns regarding applicability' (n=10)[18 21 22 29 30 33 36–38 40] (online supplemental eTable 4).

Instances categorised as 'at risk of bias' were predominantly related to case selection procedures, where enrolment criteria were inadequately described, or a significant proportion of injury video footage was not available. 'Concerns regarding applicability' largely stemmed from uncertainties surrounding rater blinding. 10 studies clearly stated whether raters were blinded to the reference standard, whereas others did not provide clear information on this aspect (table 2).

Frequency of visible signs occurrence

Studies varied in their approach to assigning visible signs to individual cases of sport-related concussion. In a greater number of studies (n=6), raters assigned multiple visible signs to each case,[21–23 27 36 39] compared with others assigning only a single visible sign (n=5).[20 24 36–38] Table 3 illustrates the frequency of occurrence of the visible signs across studies, showing considerable variability. This is with the exception of 'impact seizure',

Table 2 Characteristics of the included studies in the final review

First author, year	Study type	Sport	Competition	Sex	Number of concussions (n)	Criteria used for analysis	Rater experience	Blinded	Inter-rater reliability considered	Mechanism of injury considered	Visible signs considered
Athiviraham *et al*[17]	RCCS	BB	MLB	M	9	BCS	Expert	Yes	No	Yes (Helmet hit-by-pitch only)	No
Austin *et al*[37]	RCS	AF	NCAA Division 1	M	44	ICD	Expert	NR	No	Yes	Yes
Bruce *et al*[18]	RCS	IH	NHL	M	202	Echemendia *et al*[21] Hutchison *et al*[32]	NR	NR	No	Yes	Yes
Cassoudesalle *et al*[38]	RCS	SCR	Ligue 1, Ligue 2	M	36	ICD	Expert	NR	Yes	Yes	Yes
Clark *et al*[31]	RCS	AF	NFL	M	429	BCS	NR	NR	No	Yes	No
Cross *et al*[19]	RCCS	RU	The English Premiership, The Pro 12, The Rugby World Cup	M	182	BCS	Expert	NR	No	Yes (Tackles only)	No
Davis and Makdissi[20]	PCS	AUS	AFL	M	45	Hierarchical flow chart	Expert	Yes	No	No	Yes
Davis *et al*[8]	QQ	AUS, CK, BB, AF, IH, RL, RU	AFL, CA, MLB, NFL, NHL, NRL, WR	M	0	Questionnaire	Expert	NR	No	No	Yes
Davis *et al*[4]	QQ	AUS, CK, BB, AF, IH, RL, RU	AFL, CA, NFL, NHL, NRL, WR	M	0	Questionnaire	Expert	NR	No	No	Yes
Echemendia *et al*[21]	RCS	IH	NHL	M	202	NHL protocol	Expert	Yes	Yes	No	Yes
Elbin *et al*[22]	PCSS	AF	NFL	M	211	Makdissi and Davis[41]	Expert	NR	No	No	Yes
Gardner *et al*[39]	RCS	RL	NRL	M	20	BCS	Expert	NR	No	Yes	Yes
Gardner *et al*[23]	RCS	RL	NRL	M	60	ORAC	Expert	Yes	Yes (and intrarater reliability)	No	Yes
Gardner *et al*[40]	RCS	RL	NRL	M	25	ORAC	Expert and novice	NR	Yes	Yes	Yes
Gardner *et al*[24]	RCS	RL	NRL	M	20	BCS	Expert	NR	No	No	Yes
Gardner *et al*[25]	RCS	RL	NRL	M	60	ORAC	NR	Yes	No	Yes	Yes
Gardner *et al*[26]	RCS	RU	Super Rugby	M	34	HIA protocol	Expert	Yes	No	No	Yes
Gardner[27]	PCSS	RL	NRL	M	106	ICD	Expert	Yes	No	No	Yes
Hutchison *et al*[32]	PCS	IH	NHL	M	199	Heads-up checklist	Expert and novice	NR	Yes	Yes	No
Hutchison *et al*[33]	RCS	IH	NHL	M	197	Heads-up checklist	NR	NR	No	Yes	No
Kent *et al*[34]	RCS	AF	NCAA Division 1	M	42	BCS	Expert	NR	No	Yes	No
Lessley *et al*[35]	RCS	AF	NFL	M	367	BCS	Expert	Yes	No	Yes	No
Makdissi and Davis[28]	PCS	AUS	AFL	M	45	Makdissi and Davis[41]	Expert	Yes	Yes (and intrarater reliability)	No	Yes
Makdissi and Davis [41]	PCS	AUS	AFL	M	82	BCS	Expert	Yes	Yes	Yes	Yes
O'Halloran *et al*[29]	PCS	CK	First-Class County Championship, One-Day 50 over, T20 cricket and Men's senior national side	M	16	ICD, McCrory *et al*[45] Concussion protocols of the ICC, WR and FIFA	NR	NR	No	Yes (Helmet strikes only)	Yes
Spiegelhalter *et al*[15]	RCS	RL	WSL	F	12	Hopkinson *et al*[46]	Expert	NR	No	Yes (Tackles only)	No
Sunderland *et al*[16]	RCS	AUS	AFL, AFLW	MF	252/44	BCS	NR	NR	Yes (MOI only)	Yes	No

Continued

Table 2 Continued

First author, year	Study type	Sport	Competition	Sex	Number of concussions (n)	Criteria used for analysis	Rater experience	Blinded	Inter-rater reliability considered	Mechanism of injury considered	Visible signs considered
Suzuki *et al*[30]	RCS	RU	Collegiate competition	M	33	Suzuki *et al*[47]	NR	NR	No	Yes (Tacklers only)	No
Zuckerman *et al*[26]	RCS+PCS	AF	NFL	M	307	Hierarchical flow chart[20]	Expert and novice	NR	Yes	No	Yes

Study type coded as: RCCS, RCS, PCS, PCSS, QQ.
Sport coded as: BB, AF, IH, SCR, RU, AUS, CK, RL.
Number of concussions indicates the number of clinically diagnosed concussions.
Criteria used for analysis coded as: BCS, ICD,[8] NHL protocol, ORAC, HIA protocol.
Blinded to clinical outcome refers to; 'yes' if the study methodology actively states the rater was blinded, 'no' if the study methodology actively states the rater was not blinded and 'NR' if the study methodology does not mention either.
AF, American Football; AFL, Australian Football League; AUS, Australian Football; BB, baseball; BCS, Bespoke criteria to study; CA, Cricket Australia; CK, cricket; HIA, World Rugby Head Injury Assessment protocol; ICC, intraclass correlation coefficient ; ICD, International Consensus Definitions; IH, Ice Hockey; MLB, Major League Baseball; MOI, mechanism of injury; NCAA, National Collegiate Athletic Association; NFL, National Football League; NHL, National Hockey League; NR, not reported; NRL, National Rugby League; ORAC, Observational Review and Analysis of Concussion; PCCS, prospective cross-sectional study; PCS, prospective cohort study; QQ, Qualitative Questionnaire; RCCS, retrospective case–control study; RCS, retrospective cohort study; RL, Rugby League; RU, Rugby Union; SCR, Association Football/Soccer; WR, World Rugby.

'tonic posturing' and 'visible facial injury', which also demonstrate relatively low occurrence frequencies across the studies. Notably, on average 'slow to get up' (37.6%) and 'motor incoordination' (26.7%) were the most frequently occurring visible signs. These were followed by 'no visible sign seen' (23.6%).

Two additional studies reported the occurrence of visible signs,[26 29] but these studies took a different approach to categorisation and defining so were not included in table 3. One of these studies reported the presence or absence of visible signs rather than individual occurrences, finding visible signs in 55.9% of 34 concussion cases.[26] The others used different descriptive terms for visible signs in their cohort of 16 concussion cases, with the most frequent being 'player removes or checks helmet' (75%) and 'more than 4 s to resume' (68.8%).[29]

Visible signs: assessment of diagnostic test accuracy

The diagnostic test accuracy of individual visible signs was assessed in six studies included in this review,[18 21–23 27 28] which were chosen for their consistent and detailed definitions of visible signs. Key diagnostic accuracy measures consistently reported across these studies included sensitivity, specificity, positive predictive value and negative predictive value. Notably, the study by Elbin *et al*[22] contributed two separate data sets to the analysis, considering diagnostic accuracy measures independently for each rater involved.

Figure 2 presents the diagnostic test accuracy data in forest plots, with studies ordered based on sensitivity. Summary sensitivity and specificity values for each of the visible signs are presented in table 4. None of the studies achieved a summary sensitivity and specificity greater than 50%, precluding a comparative meta-analysis. For most visible signs, sensitivity was low (<50%) and specificity was high (>90%). The opposite was true for 'slow to get up' with a summary sensitivity of 80% and a summary specificity of 27%.

Another study examined the diagnostic test accuracy of individual visible signs; however, it used different descriptive terms and definitions and did not report all diagnostic accuracy measures, so it was not included in this analysis.[29] In that study, the visible sign with the highest sensitivity was 'player removes or checks helmet' (sensitivity 0.92, specificity 0.60) and 'more than 4 s to resume' (sensitivity 0.92, specificity 0.52).[29]

Four additional studies investigated the diagnostic test accuracy of multiple visible signs rather than individual visible signs[21 22 25 27] (online supplemental eTable 5). The limited number of studies precluded the calculation of summary sensitivity and specificity values. However, a general trend observed was that as the number of visible signs present increased, sensitivity tended to decrease while specificity increased. Similarly, positive predictive values and negative predictive values generally increased with the number of visible signs present.

Mechanism of injury characteristics

Four studies examined how characteristics of the mechanism of injury were statistically associated with a diagnosis of sport-related concussion.[18 19 25 30] Online supplemental eTable 6 outlines the mechanism of injury characteristics that either significantly increased or decreased the likelihood of a sport-related concussion diagnosis across these studies. The studies encompassed three sports—ice hockey, rugby league and rugby union—highlighting challenges in directly comparing mechanisms of injury due to sport-specific differences. Nonetheless, common themes emerged: mechanism of injury characteristics such as high speed of impact, direct head contact with a solid

Table 3 Concussion counts and visible sign types and frequencies

First author, year	Sport	Number of concussion (n)	Lying motionless (%)	Motor incoordination (%)	No protective action–floppy (%)	Impact seizure (%)	Tonic posturing (%)	Blank/ vacant look (%)	Slow to get up (%)	Clutching head/ face (%)	Visible facial injury (%)	No visible sign seen (%)
Austin *et al*[37]	AF	44	9.1	11.4	–	–	2.3	–	11.4	29.5	–	22.7
Cassoudesalle *et al*[38]	SCR	36	11.1	–	–	–	–	–	–	–	–	–
Davis and Makdissi[20]	AUS	45	4	16	25	2	–	7	22	9	11	4
Echemendia *et al*[21]	IH	202	6.4	17.3	–	–	–	2	39.6	7.9	5	53
Elbin *et al*[22]	AF	211	18.8	28.4	17.6	5.7	–	1.7	65.9	29.7	0.7	26
Gardner *et al*[39]	RL	20	30	66	45	–	–	53	–	35	–	0
Gardner *et al*[23]	RL	60	40	58.3	–	5	–	75	100	63.3	–	0
Gardner *et al*[40]	RL	20	55	–	–	–	–	–	–	–	–	20
Gardner[27]	RL	106	20.8	36.8	20.8	2.8	5.7	53.8	–	–	–	43.4
Zuckerman *et al*[36] (novice raters)	AF	96	9.5	15	5.5	0	–	0	21.5	4.5	0	43.5
Zuckerman *et al*[36] (expert raters)	AF	211	18.5	28.5	17.5	6	–	1.5	66	29.5	0.5	26.5
Average percentage % (range %)			21.2 (4–55)	26.7 (11.4–66)	19.1 (5.5–45)	2.9 (0–6)	5.3 (2.3–5.7)	20.5 (0–75)	37.6 (11.4–100)	23.4 (4.5–63.3)	2.2 (0–11)	23.6 (4–53)

Sport coded as: AF, IH, SCR, AUS, RL.
Number of concussions indicates the number of clinically diagnosed concussions.
Studies where the visible sign was either not considered or not reported is denoted by '–'.
AF, American Football; AUS, Australian Football; IH, Ice Hockey; RL, Rugby League; SCR, Association Football/Soccer.

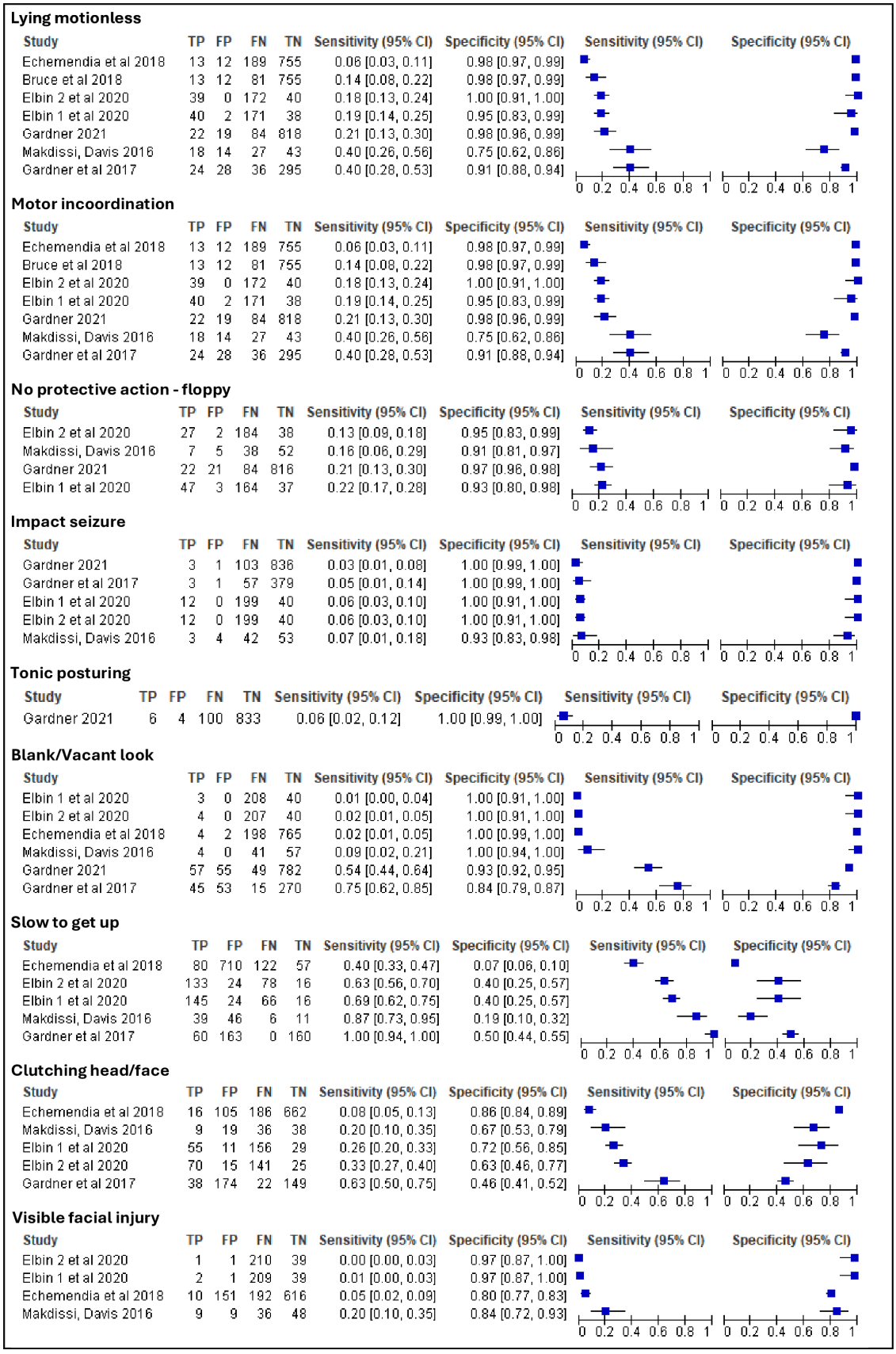


Figure 2 Forest plots displaying study sensitivity and specificity of individual visible signs. FP, false positive; FN, false negative; TN, true negative; TP, true positive.

Table 4 Sensitivity and specificity of the visible signs

Visible signs	Studies (n)	SRC cases/total number of cases	Summary sensitivity (95% CI)	Summary specificity (95% CI)
Lying motionless	7	929/3760	0.20 (0.13 to 0.30)	0.96 (0.92 to 0.98)
Motor incoordination	7	929/3760	0.32 (0.24 to 0.42)	0.91 (0.87 to 0.94)
No protective action–floppy	4	573/1547	0.18 (0.13 to 0.23)	0.96 (0.91 to 0.98)
Impact seizure	5	633/1987	0.05 (0.04 to 0.07)	1.00 (0.97 to 1.00)
Blank/vacant look	6	835/2899	0.10 (0.02 to 0.39)	0.99 (0.95 to 1.00)
Slow to get up	5	729/1956	0.80 (0.46 to 0.95)	0.27 (0.14 to 0.45)
Clutching head/face	5	729/1956	0.27 (0.14 to 0.46)	0.69 (0.55 to 0.80)
Visible facial injury	4	669/1573	0.03 (0.01 to 0.12)	0.92 (0.77 to 0.97)

SRC, sport-related concussion.

object and foul play consistently increased the likelihood of sport-related concussion. Conversely, lower speed and reduced overall force of impact were associated with decreased likelihood of sport-related concussion.

In addition, two studies focused on the frequency of mechanism of injury characteristics, both involving female athletes.[15 16] In women's rugby league, sport-related concussion incidence was higher among tacklers (3.5/1000 head contacts, 95% CI 0.9 to 9.5) and during knee-to-head contact (25.6/1000 knee-to-head contact events, 95% CI 16.7 to 37.7), with minimal influence from team standard or time in the match.[15] Comparatively, in Australian Football competitions, female athletes had higher sport-related concussion rates following actions like bumping (incidence rate ratio (IRR)=8.11, 95 % CI 1.36 to 48.53), being bumped (IRR=4.63, 95 % CI 2.05 to 10.46) or being tackled (IRR=2.66, 95 % CI 1.17 to 6.03).[16] Female athletes also showed increased sport-related concussion rates following contact with the ground (IRR=2.79, 95% CI 1.45 to 5.37), opposing player's head (IRR=3.29, 95% CI 1.64 to 6.61) or pelvis/hip (IRR=5.53, 95% CI 1.92 to 15.91).[16]

Reliability

Six studies examined the inter-rater reliability of individual visible sign identification[21 23 28 36 40 41] (online supplemental eTable 7). Among these, two studies explored agreement between novice and expert raters,[36 40] while another two focused on intrarater reliability.[23 28] Cohen's kappa coefficients (K) were most frequently chosen as the statistical measure of inter-rater reliability (n=4).[23 36 40 41] Across these studies, experts generally demonstrated higher levels of agreement compared with novices.[36 40] However, variability in performance was observed, with K values for novice raters ranging from 0.44 (weak) to 0.68 (moderate), and for expert raters ranging from 0.08 (none) to 0.89 (strong). Notably, one study[41] significantly influenced expert rater variability, showing minimal to no agreement across all visible signs using the McHugh K agreement classification.[42]

Higher inter-rater agreement was consistently observed for visible signs such as 'lying motionless', 'slow to get up' and 'clutching head/face', while 'blank/vacant look' exhibited lower agreement levels. One study exclusively reported inter-rater reliability for mechanism of injury characteristics, indicating 'almost perfect' agreement (K=0.95, 95% CI 0.83 to 1.00).[16]

Intrarater reliability was assessed for individual visible signs among expert raters only.[23 28] In Australian Football, the intraclass correlation coefficient (ICC) demonstrated varied agreement levels, with 'visible facial injury' showing the highest ICC (rater 1=1.00, rater 2=0.93) and 'blank/vacant' look the lowest (rater 1=0.63, rater 2=0.36).[28] In rugby league, K demonstrated moderate to perfect intrarater reliability for all visible signs identified by two expert raters, with 'impact seizure', 'slow to get up' and 'clutching head/face' showing perfect agreement.[23] The visible sign with the lowest level of intrarater agreement, although still moderate, was 'blank/vacant look'.[23]

DISCUSSION

The integration of video review into protocols for identifying sport-related concussion has become integral across major international sporting organisations.[4] However, variations in definitions for individual visible signs among protocols across different sports were noted, encompassing a total of 19 distinct visible signs. The only universally recognised signs across all protocols were 'loss of consciousness' or 'lying motionless' and 'motor incoordination'.[4] In many sports, the identification of these signs on video mandates permanent removal of the athlete from play (eg, Australian Football League, World Rugby, National Football League, Cricket Australia, National Rugby League), while in others, such as the National Hockey League, removal for further assessment is required.[4] Despite many sports using the term 'loss of consciousness', it is recognised that a clinical assessment is required to diagnose this sign and cannot be done on video review alone. The preferred term therefore when performing video review is 'lying motionless'.[4]

This systematic review represents the first comprehensive assessment of the scientific evidence investigating the role of video review in sport-related concussion identification. The increasing adoption of this technology in professional sports organisations' protocols underscores its importance in guiding clinicians' decision-making regarding athlete management on the field of play. Therefore, this review will in part help to inform the application of video review. With many of the studies identified being retrospective in design, it is recognised that future studies exploring real-time use are required to better inform practice.

Despite growing scientific and clinical interest in sport-related concussion over recent decades, our review identified a relatively modest total of 29 studies. Most of these studies were retrospective and included professional athletes. This number contrasts with 98 studies included in a review focusing on more advanced technologies for sport-related concussion assessment.[43] Only two studies included female athletes,[15 16] and sample sizes across studies were generally small, with heterogeneous criteria for video review. Given the critical role of video review in current protocols and clinical decision-making, we expected a larger and more consistent evidence base underpinning its application. The size of the evidence base may reflect limited access to video review technology due to resource constraints in research and applied settings, outside of professional competitions. Additionally, anonymising the injury video footage is challenging, introducing further ethical considerations for researchers.

Concerns regarding bias and applicability were notable, with only 28% of studies considered to have low risk of bias. The remaining 72% were either at high risk of bias or had concerns regarding applicability. This raises questions about the generalisability of findings across different sport settings and for female athletes, who may be at higher risk of concussion but are underrepresented in research.[44] Addressing methodological issues such as including all consecutive cases, studies conducted in both sexes, ensuring rater blinding, performing real-time review and adopting international visible signs consensus definitions to standardise assessments is important for future research efforts to overcome deficiencies currently present and establish a more homogenous evidence base.[8] Future studies should also consider how adopting this technology and definitions impacts clinical practice and outcomes. Detailed comment on para-sport participants goes beyond the scope of this review, but it is important that dedicated studies of this athlete group are completed as this body of literature develops.

In diagnosing sport-related concussion, athletes may disclose symptoms, display overt visible signs or exhibit findings on examination, or a combination thereof, following a direct or indirect head impact. Historically, loss of consciousness was pivotal in confirming sport-related concussion diagnoses, but a broader range of visible signs is now recognised.[8] Recent studies indicate that loss of consciousness occurs in less than 10% of cases,[3] yet our review suggests a higher occurrence of lying motionless in professional male settings (average 21.2%, range 4%–55%). However, the most frequent visible signs identified were 'slow to get up' and 'motor incoordination'. Notably, 'no visible sign seen' occurred in 23.6% of cases, and with the non-specific nature of 'slow to get up', this highlights that video review cannot be solely relied on to inform clinical decision-making and should be supported by a multimodal sideline assessment.[1 2] Factors contributing to variations in the presence of visible signs across studies and sports may include helmet use, mechanisms of injury variations across sports and potential strategic advantages in sports for remaining on the ground.[23]

Examining the diagnostic test accuracy of individual visible signs, our review identified only six studies suitable for direct comparison due to inconsistent definitions of visible signs.[18 21–23 27 28] Overall, no single visible sign demonstrated both high sensitivity and specificity, indicating no clinically useful standalone test for the diagnosis of sport-related concussion. Most visible signs had a low sensitivity (<50%) and a high specificity (>90%), suggesting their utility in prompting athlete removal while cautioning against ruling out sport-related concussion in their absence. 'Slow to get up', however, showed relatively high sensitivity (80%) and low specificity (27%), suggesting its usefulness in identifying potential sport-related concussion cases following a significant direct or indirect head impact, and the need for athletes to receive comprehensive multimodal sideline assessment.

Even fewer studies considered combined visible signs to diagnose sport-related concussion. Those that did demonstrated that as the number of visible signs increased, sensitivity decreased while specificity, positive predictive value and negative predictive value increased.[21 22 25 27] Future studies should explore the role of mechanism of injury characteristics alone or in combination with visible signs to enhance diagnostic accuracy or help identify athletes that require subsequent clinical assessment. Four studies examined how mechanism of injury characteristics alone may influence the likelihood of diagnosing sport-related concussion.[18 19 25 30] The applicability of these study results across different sports may be limited, as two studies focused exclusively on specific sports-related mechanisms of injury.[19 30] Overall, mechanisms of injury that increase or decrease the likelihood of a subsequent sport-related concussion diagnosis are related to the force or speed of impact. It is evident that mechanisms of injury differ between sexes;[16] therefore, future sport-specific and sex-specific research is warranted when studying mechanisms of injury.

An inherent limitation of visible signs is that they are subject to human error, as evidenced by studies examining inter-rater reliability.[21 23 28 36 40 41] Variability in reliability may not solely stem from the performance of the individual raters. Contributing factors include the quality of available video footage, including both the visual quality and number of camera angles, and the precision of visible signs definitions.[41] Recommendations for sideline medical personnel include access to real-time, multiangle camera feeds including broadcast footage for accurate review, with the ability to replay at various speeds.[4] International consensus definitions may increase precision if widely adopted, as they facilitate a degree of standardisation which is important as the clinical diagnosis is the same regardless of setting.[8] Using expert raters appears beneficial, given their demonstrated higher reliability,[36 40] with many sports organisations also requiring these individuals to be experienced doctors.[4] It could, therefore, be advantageous for individuals to demonstrate both characteristics. Studies on intrarater reliability have shown acceptable levels,[23 28] suggesting that while individual reliability may be adequate, ongoing education in visible signs identification could enhance consistency across raters. Several international sporting organisations (National Football League, National Hockey League, National Rugby League, World Rugby) already offer specific educational modules for this purpose.[4]

Recommendations for future research

We propose several recommendations for future research aimed at addressing the current limitations in the evidence base and enhancing the use of video review in sport-related concussion identification within organisational protocols:

1. Enrolment of larger and representative samples: Include larger and more diverse samples, encompassing female athletes, a broader spectrum of sports and various competition levels.
2. Prospective, blinded study designs: Employ prospective, consecutive study designs where raters are blinded to the clinical outcome.
3. Investigate the impact on clinical practice of standardised visible signs definitions: Possibly adopting the international consensus definitions for this purpose.[8]
4. Exploration of cross-sport patterns: Investigate whether there are overarching patterns or mechanistic explanations for the variability in visible signs occurrence across different sports.
5. Diagnostic test accuracy studies: Conduct diagnostic accuracy studies on both individual and combined visible signs across a wider array of sports to inform decisions regarding mandatory athlete removal or further assessment.
6. Sport-specific mechanism of injury characteristics: Gain a better understanding of sport-specific mechanism of injury characteristics and their role in risk-prediction models.
7. Improving inter-rater reliability: Explore the reasons behind variations in inter-rater reliability and evaluate interventions aimed at enhancing reliability. This could include stepwise video review process guidance or educational programmes.

Study limitations

Several limitations should be considered regarding this review. Eligibility criteria may have inadvertently excluded relevant studies, such as those not accessible in English or without full-text access. This review does not encompass evidence specific to individuals under 18 years old or participants in para-sports, and only two studies included female athletes. Despite efforts to standardise definitions, subtle variations across studies introduce potential errors when calculating summary sensitivity and specificity values. Finally, potential publication bias may influence overall results and conclusions.

CONCLUSIONS

This systematic review aims to enhance the understanding of the role of video review in sport-related concussion identification and its optimal integration into sporting organisations' management protocols. The results support the use of expert-led video review for visible signs identification. However, given its limited diagnostic accuracy, this approach should complement a comprehensive sideline multimodal assessment. Further research should address the current methodological limitations to improve generalisability, explore standardised visible signs definitions and the potential benefits of incorporating the mechanism of injury characteristics.

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